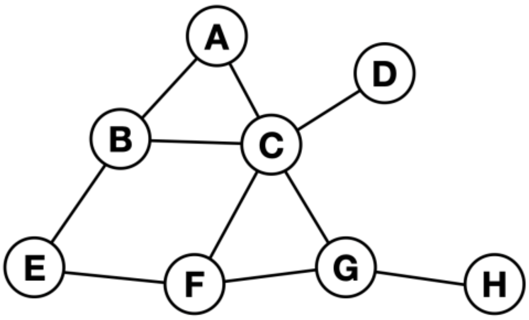
Multimodal Deep Learning

Answer Sheet 5

Date: June 17, 2025

Exercise 1: Graph representations



Given is above's graph with the vertices $\mathcal{V} = \{$**A, B, C, D, E, F, G, H**$\}$.

(a) Write down its representation as list of edges $\mathcal{E}$.
Answer: $\mathcal{E} = \{$**(A,B), (A,C), (B,E), (B,C), (C,D), (C,F), (F,G), (G,H)**$\}$

(b) Write down its representation as the adjacency matrix $\mathcal{A}$.
Answer:

$$\mathcal{A} = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Exercise 2: Graph scaling

Assume a social graph of 1,000,000 members (nodes) and that each node is connected to 200 randomly selected other nodes.

(a) How large is the adjacency matrix of this graph?
Answer: The size of the adjacency matrix $\mathcal{A}$ is $(10^6, 10^6)$, which contains $10^6 \times 10^6 = 10^12$ elements – a very sparse matrix.

(b) How many entries of the adjacency matrix are non-zero?
Answer: There will be only $200 \times 10^6 = 2 \times 10^8$ entries, which is equivalent to 0.02%.

(c) Given a random node, estimate how many hops are required until 50% of the network can be reached?
Answer: With Erdős-Rényi model, a.k.a random graph theory, given $k = 200$ and h is the number of hops. Hence,

$$R(h) = 1 + \sum_{i=0}^{h-1} k(k-1)^i \approx 1 + k \cdot \frac{(k-1)^h - 1}{k-2}$$

As $R(h) = 50\%$ of 1,000,000.
Therefore,

$$500000 = 1 + 200 \times \frac{199^h - 1}{198}$$
$$\therefore \quad h \approx 2.48 \approx 2.5 \text{ hops}$$

(d) Why may this estimation not hold in real-world social graphs?

(e) How would above's scaling behavior affect the application of graph convolutional neural networks?